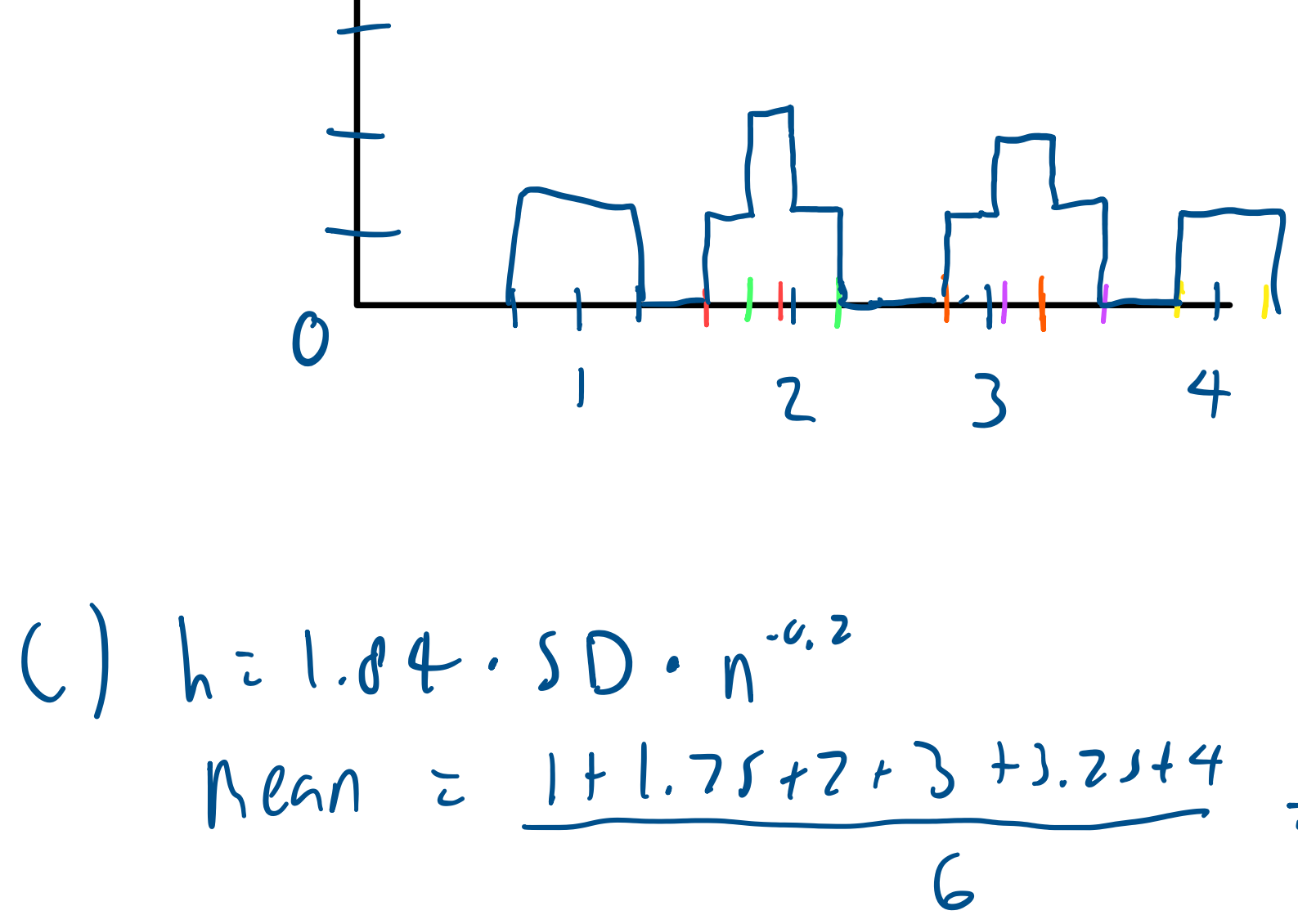
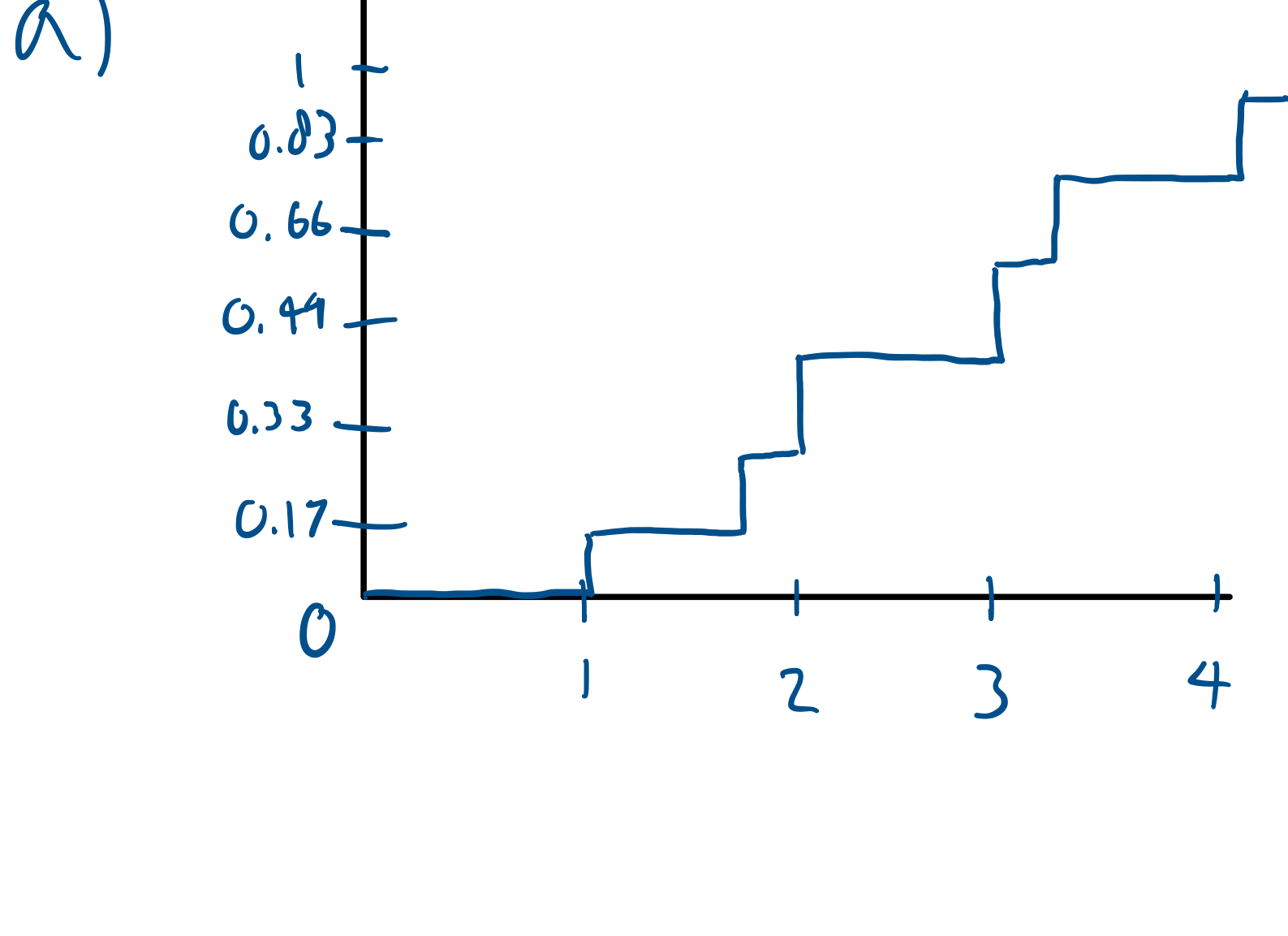


Problem 1: $[1, 1.75, 2, 3, 3.25, 4]$



c)

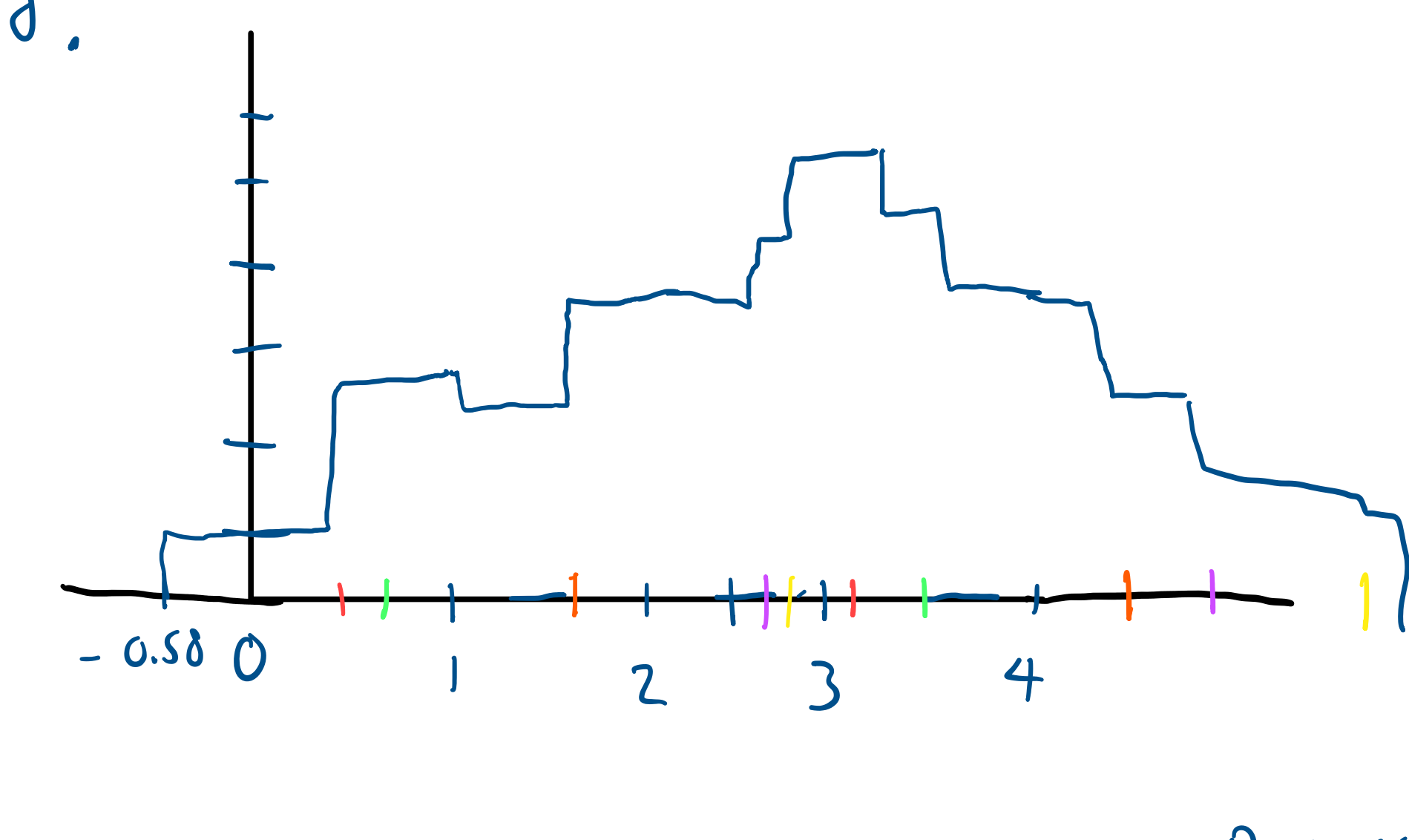
$$h = 1.84 \cdot SD \cdot n^{-0.2}$$

$$\text{mean} = \frac{1 + 1.75 + 2 + 3 + 3.25 + 4}{6} = 2.5$$

$$\sum (x - \bar{x})^2 = 2.25 + 0.5625 + 0.25 + 0.25 + 0.5625 + 2.25 = 6.125$$

$$SD = \sqrt{\frac{6.125}{5}} = 1.107$$

$$\text{Silverman's bandwidth} = 1.84 \cdot 1.107 \cdot 6^{-0.2} = 1.42$$



e. B and D are different in that D overlaps over the entire range and B has areas with 0 density. B also goes beyond the range of the data, whereas D does not.

Problem 2: $X = [-1, -0.5, 0, 0.5, 1]$

a) $X=0, h=0.6, n=5 \rightarrow 0, \underbrace{1, 1, 1, 0}_3$

$$\hat{f}_h(0) = \frac{3}{2(5)(0.6)} = \frac{3}{6} = 0.5$$

b) $E[\hat{f}_h(0)] = \frac{1}{2(0.6)} \cdot P(-0.6 \leq X \leq 0.6)$

$$= \frac{1}{1.2} \cdot (0.7257 - 0.2743) = \frac{0.4514}{1.2} = 0.376$$

c) $f(0) = \frac{1}{\sqrt{2\pi}} e^{0/2} = \frac{1}{\sqrt{2\pi}} = 0.399$

d) b and c are close but not equal. Since the estimate is smaller than the true bias, there is negative bias.

e) $\frac{1}{2(0.1)} \cdot P(-0.1 < X < 0.1)$

$$= \frac{1}{0.2} \cdot (0.5398 - 0.4602)$$

$$= 5 \cdot (0.0796)$$

$$= 0.398$$

As h shrinks, the gap also shrinks, indicating that the expected KDE value is closer to the actual KDE value.

f) $\hat{f}_h$ alone is not a consistent estimator because its accuracy is reliant on the size of h . If h is chosen appropriately it can be a consistent estimator

g) The main considerations for choosing bandwidth is bias. If the bias is large, shrinking the bandwidth can help find the true KDE

3. $[5, 7, 7, 8, 10]$ $h=1.5$

a. $\hat{f}_h(x) = \frac{1}{2nh} \sum_{i=1}^n \mathbb{I}(|x_i - x| < h)$

$$x=6: \frac{3}{15} = 0.2$$

$$x=6.5: \frac{2}{15} = 0.133$$

$$x=7: \frac{3}{15} = 0.2$$

$$x=8: \frac{2}{15} = 0.133$$

$$x=8.5: \frac{1}{15} = 0.067$$

b. $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x_i - x}{h}\right)^2\right)$

$$= \frac{1}{7.5} \sum_{i=1}^5 \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{x_i - x}{1.5}\right)^2\right]$$

$$\hat{f}_h(6) = \frac{1}{7.5} (0.3144 + 0.3144 + 0.3144 + 0.1640 + 0.0114) = 0.151$$

$$\hat{f}_h(6.5) = \frac{1}{7.5} (0.242 + 0.3774 + 0.3774 + 0.242 + 0.0262) = 0.1687$$

$$\hat{f}_h(7) = \frac{1}{7.5} (0.164 + 0.3989 + 0.3989 + 0.3144 + 0.0540) = 0.1780$$

$$\hat{f}_h(8) = \frac{1}{7.5} (0.054 + 0.3144 + 0.3144 + 0.3989 + 0.1640) = 0.1674$$

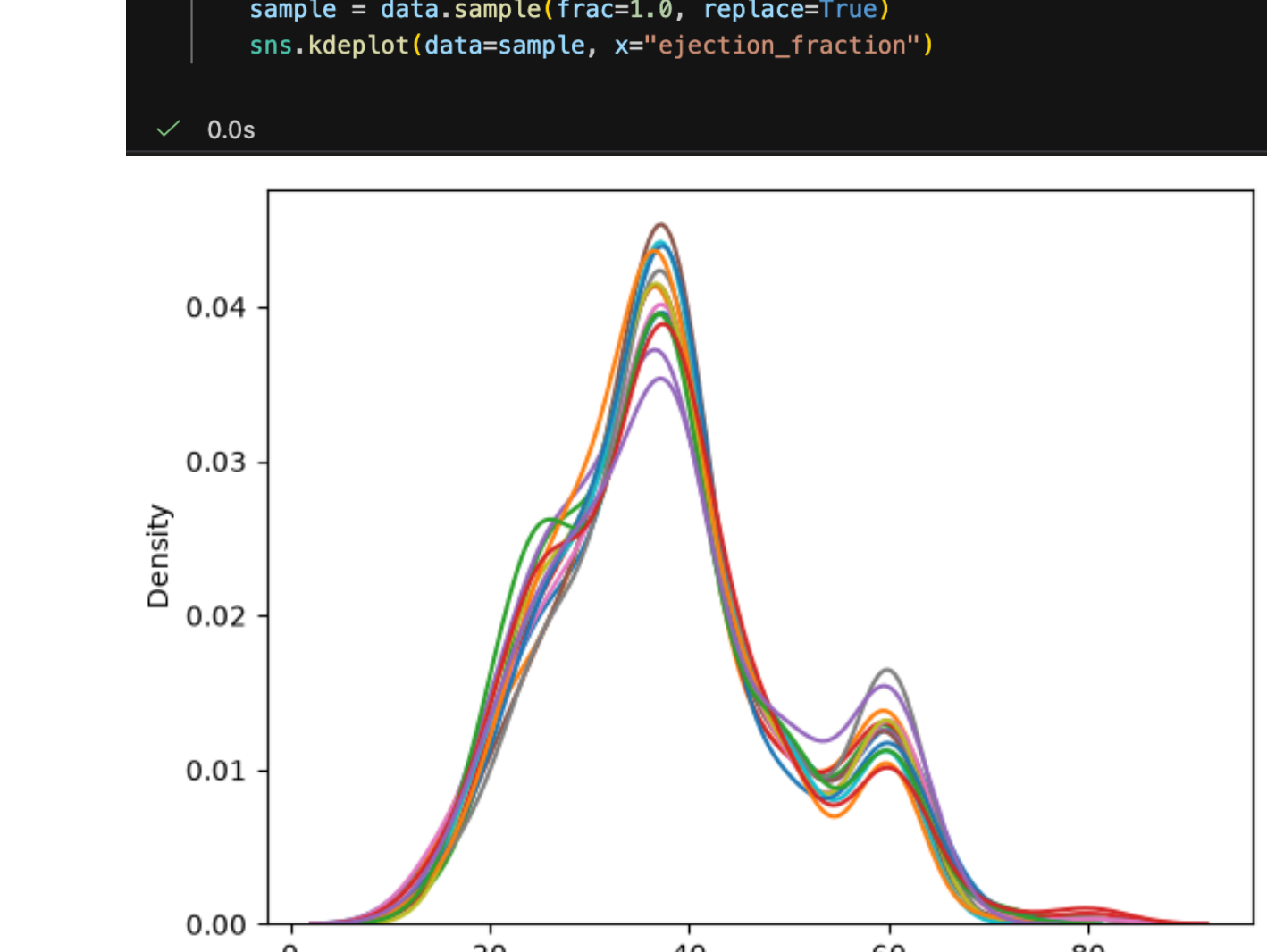
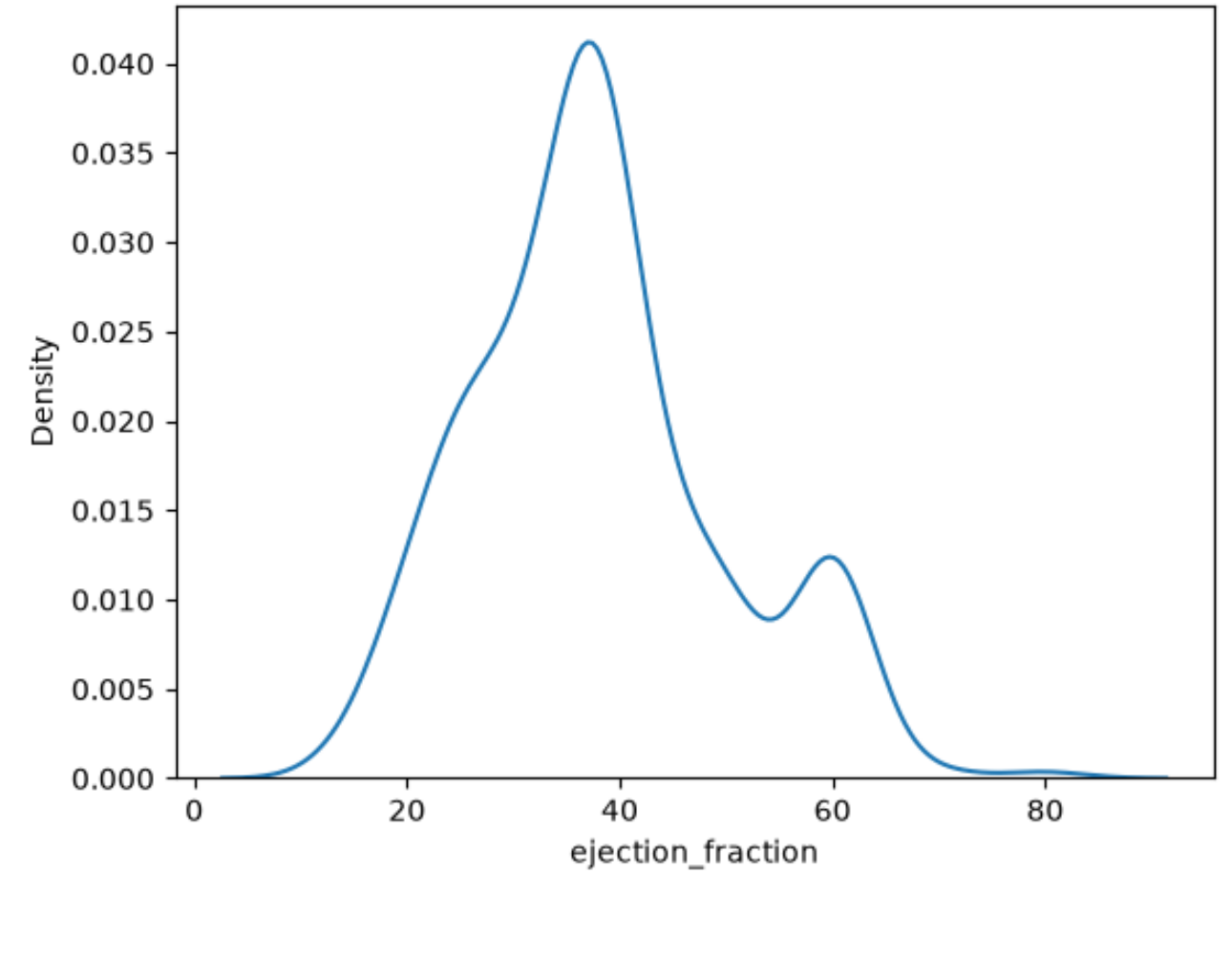
$$\hat{f}_h(8.5) = \frac{1}{7.5} (0.0262 + 0.242 + 0.242 + 0.3774 + 0.242) = 0.1506$$

c. For a, the biggest jump (-0.133) occurs between $x=8$ and $x=8.5$. This happens because there is only one point being included in the indicator versus three at $x=8$. The Gaussian case has no specific values required to be calculated, so there are no sudden jumps.

d. In practice, the uniform kernel could be preferred for simplicity, but at times there are hard jumps. In these instances, the Gaussian kernel may be preferred.

3.

a.



c. The values that appear to be most reliable appear to be in the 40-50 range, as there is the least amount of variation (gaps) between plots. The 65-80 range also appears reliable. There is noticeable variation in the 20-40 and 50-60 range.